

Tues (Me)

Thurs (Jaehoon)

Part 1

Non-bipartite Turán Problem

Puzzle: we are allowed to pick n irrational numbers $(x_1, \dots, x_n)$, we want to maximise

rational pairs (x_i, x_j) (if $x_i + x_j \in \mathbb{Q}$)

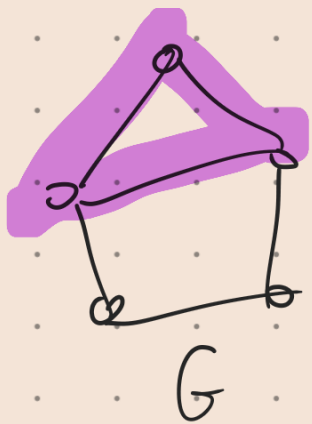
Problem (Turán-type) How many edges we can have without having another (given) graph as a subgraph?

Def: G contain a copy of H if

$\exists$ injective map $\varphi: V(H) \rightarrow V(G)$

that preserves the adjacency

if $u, v \in E(H) \Rightarrow \varphi(u), \varphi(v) \in E(G)$



H

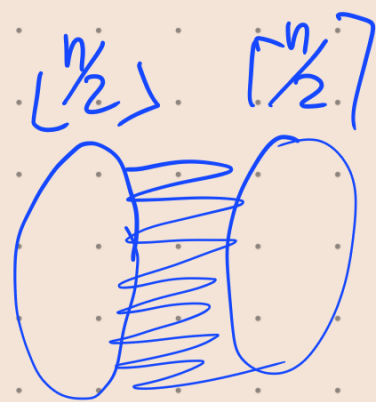
Def: extremal number of H

$$ex(n, H) = \max \# \text{ edges in } G \left\{ \begin{array}{l} |G| = n \\ H\text{-free} \end{array} \right.$$

Thm 1.2 (Mantel 1907) $\forall$ n -vx G Δ -free

$$\Rightarrow e(G) \leq \lfloor \frac{n^2}{4} \rfloor$$

$$ex(n, K_3) = \lfloor \frac{n^2}{4} \rfloor$$



Ex Solve the rational pair puzzle.

Ex Prove that $\forall$ tree T with max-deg k

$$\Delta(T) \leq k, \Rightarrow ex(n, T) \leq kn$$

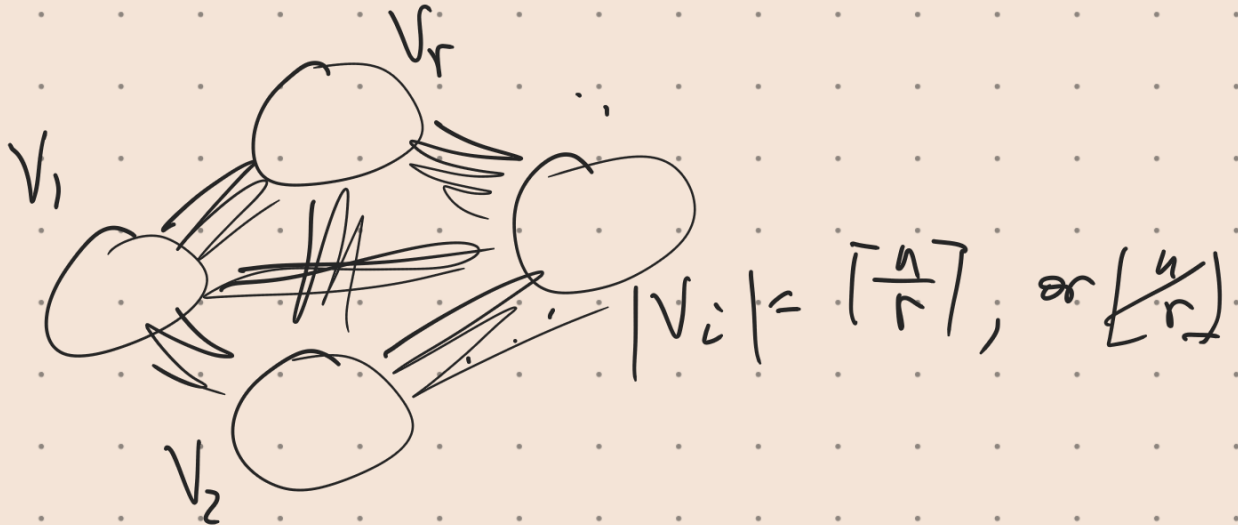
Problem 1.5 (Extremal Structure/Stability)

How do H -free extremal graphs look like?

If we have an almost maximiser, then does it look similar to the extremal graph?

Def: r -partite Turán graph. Take $n \in \mathbb{N}$
 $T_r(n)$ = balanced complete r -partite graph

vxs



- The largest clique in $T_r(n)$ is K_r .

Thm 1.6 (Turán 41) $\forall r \geq 2, \forall n$ -vx G

$$K_{r+1}\text{-free} \implies e(G) \leq e(T_r(n)) = \text{ex}(n, K_{r+1}) \\ = \left(1 - \frac{1}{r}\right) \binom{n}{2} - O(r)$$

Thm 1.7 (Erdős-Simonovits Stability 66)

Let $\epsilon > 0, \exists \delta > 0$ s.t. the following

holds. Let G be an n -vx K_{r+1} -free graph.

If $e(G) \geq ex(n, K_{r+1}) - \delta n^2$

$\Rightarrow$ then $|E(G) \Delta E(T_r(n))| \leq \varepsilon n^2$.

(i.e. we can change $G \rightsquigarrow T_r(n)$
by changing $\leq \varepsilon n^2$ adjacencies)

Recall :

• $\vec{v} \in \mathbb{R}^k$, ℓ_p -norm $\|\vec{v}\|_p = \left(\sum_{i \in [k]} v_i^p \right)^{1/p}$.

• G graph, write A_G = adjacency matrix

$|G| = n$, A_G $n \times n$ matrix

$$a_{ij} = \begin{cases} 0 & \text{if } v_i \not\sim v_j \\ 1 & \text{if } v_i \sim v_j \end{cases}$$

$G = K_4$

$$A_{K_4} = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

- For $k \times k$ symm. matrix A ,

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$$

Frobenius norm (Hilbert-Schmidt norm)

$$\begin{aligned} \|A\|_F &= \left(\sum_{i,j \in [k]} a_{ij}^2 \right)^{1/2} = \sqrt{\text{tr}(A^2)} \\ &= \sqrt{\lambda_1^2 + \lambda_2^2 + \dots + \lambda_k^2} \end{aligned}$$

§ 1.2 Symmetrisation à la Motzkin - Straus

- n -vx graph G , $\vec{x} \in \mathbb{R}^n$, define

$$f_G(\vec{x}) = \vec{x}^T A_G \vec{x} = \sum_{i,j \in [n]: v_i v_j \in E(G)} x_i x_j$$

$$\left(f_G : \mathbb{R}^n \rightarrow \mathbb{R} \right) \quad \vec{x} = (x_1, \dots, x_i, \dots, x_n) \in \mathbb{R}^n$$

- Def a simplex

$$S_n = \{ \vec{x} \in \mathbb{R}^n : x_i \geq 0 \forall i, \sum x_i = 1 \}$$

Rmk: Think of $\vec{x} \in S_n$ as a weight
funct. on $V(G)$.

$\vec{x} \rightsquigarrow$ weighted subgraph $G_{\vec{x}} \subseteq G$

$$f_G(\vec{x}) = \vec{x}^T A_G \vec{x} = 2 \cdot e(G_{\vec{x}})$$

In particular, take $\vec{x} = (\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n})$

$$f_G(\vec{x}) = \frac{2}{n^2} \cdot e(G)$$

clique number

Thm 2.1 (Motzkin - Straus)

Given n -vx G with $\omega(G) = k$ (K_{k+1} -free)

$\vec{x} \in S_n$
 $\implies \exists \vec{y} \in S_n$ s.t. $\begin{cases} \cdot f_G(\vec{y}) \geq f_G(\vec{x}) \\ \cdot \text{supp}(\vec{y}) = K_k \end{cases}$

In particular,

$$f_G(\vec{x}) \leq \frac{k-1}{k}, \quad \forall \vec{x} \in S_n.$$

Idea : "mass transportation"

if $\exists$ two non-adjacent u, v in support,
then moving mass from one to another
will $\begin{cases} \bullet \text{ not decrease } f_G(\cdot) \\ \bullet \text{ decrease support.} \end{cases}$

Pf : Take $\vec{y} \in S_n$ $\vec{y} = (y_1, \dots, y_n)$ s.t.

- $f_G(\vec{y}) \geq f_G(\vec{x})$
- subj to above, minimal support

Suppose $\text{supp}(\vec{y}) \supseteq \{v_1, v_2\}$ s.t. $v_1 \not\sim v_2$.

Take $\forall \vec{z} = (z_1, z_2, 0, \dots, 0)^T$

$\Rightarrow$ as $v_1 \not\sim v_2 \Rightarrow \boxed{z^T A_G z = 0}$

$\bullet$ Then $\forall \alpha \in \mathbb{R}$ $f_G(\vec{y} + \alpha \vec{z}) = (\vec{y} + \alpha \vec{z})^T A (\vec{y} + \alpha \vec{z})$

$$= \underbrace{y^T A y}_{f_G(y)} + \underbrace{\alpha z^T A y}_{\uparrow \text{ same } \uparrow} + \underbrace{\alpha y^T A z}_{\uparrow \text{ same } \uparrow} + \boxed{\alpha^2 z^T A z}$$

$$\Rightarrow f_G(\vec{y} + \alpha \vec{z}) = f_G(\vec{y}) + \underline{a^T \cdot z}, \text{ where}$$

$$\vec{a} = 2\alpha A \cdot \vec{y}$$

$$\vec{a} = (a_1, a_2, \dots, a_n)$$

$$a^T \cdot z = a_1 z_1 - a_2 z_1$$

$$= z_1 (a_1 - a_2)$$

$$\geq 0$$

$$\bullet \vec{y} \in S_n, \quad \sum y_i = 1$$

$$\text{WANT } \underline{\vec{y} + \alpha \vec{z} \in S_n} \Rightarrow \sum z_i = 0$$



$$\text{then } \Rightarrow \text{ take } z = (z_1, -z_1, 0, 0, \dots, 0)$$

$$\bullet \text{ by choosing } z_1 \text{ w./ appropriate sign}$$

$$\text{Write } \vec{y}' = \vec{y} + \alpha \vec{z}$$

$$\left\{ \begin{array}{l} f_G(\vec{y}') \geq f_G(\vec{y}) \geq f_G(\vec{x}) \end{array} \right.$$

$$\left\{ \begin{array}{l} \text{Choose } \alpha \Rightarrow \text{supp}(\vec{y}') = \text{supp}(\vec{y}) \setminus \{i\} \\ \text{for some } i \in [2] \end{array} \right.$$

$$\vec{y}' = (\underline{y_1 + \alpha z_1}, \underline{y_2 - \alpha z_1}, \dots, \dots)$$



Ex 2.2 Prove "In particular" part of Thm 2.1.

Multi-color version by Füredi - Maleki

Ex 2.3 Let G be an n -vx graph.

Take $G_1, G_2 \subseteq G$ & take $\vec{x} \in S_n$

$$\Rightarrow \exists \vec{y} \in S_n \text{ s.t. } \begin{cases} \bullet f_{G_i}(\vec{y}) \geq f_{G_i}(\vec{x}) \\ \bullet \text{supp}(\vec{y}) =: K \end{cases} \quad \forall i \in [2]$$

$$\text{indep. number } \alpha(G[K]) \leq 2$$

§ A quick application

A symm $n \times n$ matrix, $\lambda_1 \geq \dots \geq \lambda_n$

$$\bullet \|A\|_F^2 = \text{tr}(A^2) = \lambda_1^2 + \lambda_2^2 + \dots + \lambda_n^2$$

$$\Rightarrow \lambda_1^2 \leq \|A\|_F^2 \quad \text{clique number}$$

Ex 2.4 Let G be a graph w/ $\omega(G) = k$

& let $A = A_G$, then

$$\lambda_1(A)^2 \leq \frac{k-1}{k} \|A\|_F^2$$

Rmk : The bound is sharp for cliques.

$$G = K_k \rightarrow \begin{cases} \lambda_1(K_k) = k-1 \\ \|A_{K_k}\|_F = \sqrt{k^2 - k} \end{cases}$$

So far, we know $ex(n, K_{n+1})$
 $ex(n, H)$

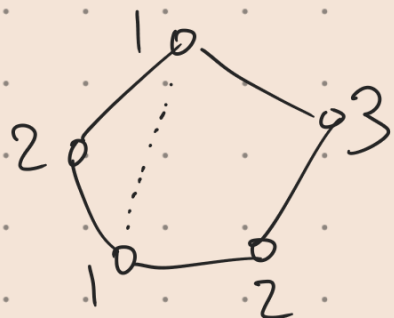
What about generic H ?

§ 3 Erdős - Simonovits - Stone Thm

Def Given H , its **chromatic number**

$\chi(H)$ is the minimum # colours needed
 so that $V(H)$ can be coloured so that

$\forall u \sim v$, u, v receive different colours.



$$\chi(C_5) \leq 3$$

" = " $\uparrow$

Thm 3.1 Let H be a graph, then

$$ex(n, H) = \left(1 - \frac{1}{\chi(H)-1} + o(1) \right) \frac{n^2}{2}$$

($o(1) \rightarrow 0$ as $n \rightarrow \infty$)



Rmk 1) if $\chi(H) \geq 3$, i.e. H is not bipartite, then ESS solves $ex(n, H)$.

But if $\chi(H) = 2$, i.e. H bipartite, then

$$ex(n, H) = o(n^2) \quad \hookrightarrow \exists c = c_H$$

In fact, $ex(n, H) = O(n^{2-c}).$

Next time

, prove - thm 3.1

- Stability method

⋮